

DATE

22/05/2025

MULTI-OBJECTIVE REINFORCEMENT LEARNING FOR SUSTAINABLE INVESTMENT STRATEGIES USING FINRL

REINFORCEMENT LEARNING 2025-2

PRESENTED BY

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FINAL PROJECT

Daniel R.M.

01

MOTIVATION

MOTIVATION I

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Investing wisely is no longer just about earning more, but
also about taking less risk and being more responsible

- Traditional RL in finance focuses only on maximizing returns.
- But in real-world investing, people care about more than just profit.
- This project uses MORL to consider three key objectives:
 - r_1 – Return: Grow the portfolio by maximizing profit.
 - r_2 – Risk: Penalize high volatility to reduce exposure.
 - r_3 – ESG: Encourage investments aligned with ethical and sustainable values.

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OBJECTIVES

OBJECTIVE

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Design and evaluate a multi-objective reinforcement learning system capable of generating investment strategies that balance financial return, risk, and sustainability

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METHODOLOGY

- Analyze FinRL's trading environment and **PPO** baseline.
- Build a diversified portfolio and assign **ESG** scores.
- Redesign the reward function as a 3-part vector: **return**, **risk**, and **ESG**.
- Apply **linear scalarization** to combine the objectives into a single reward.
- Train agents using PPO with both manual and evolved preferences.

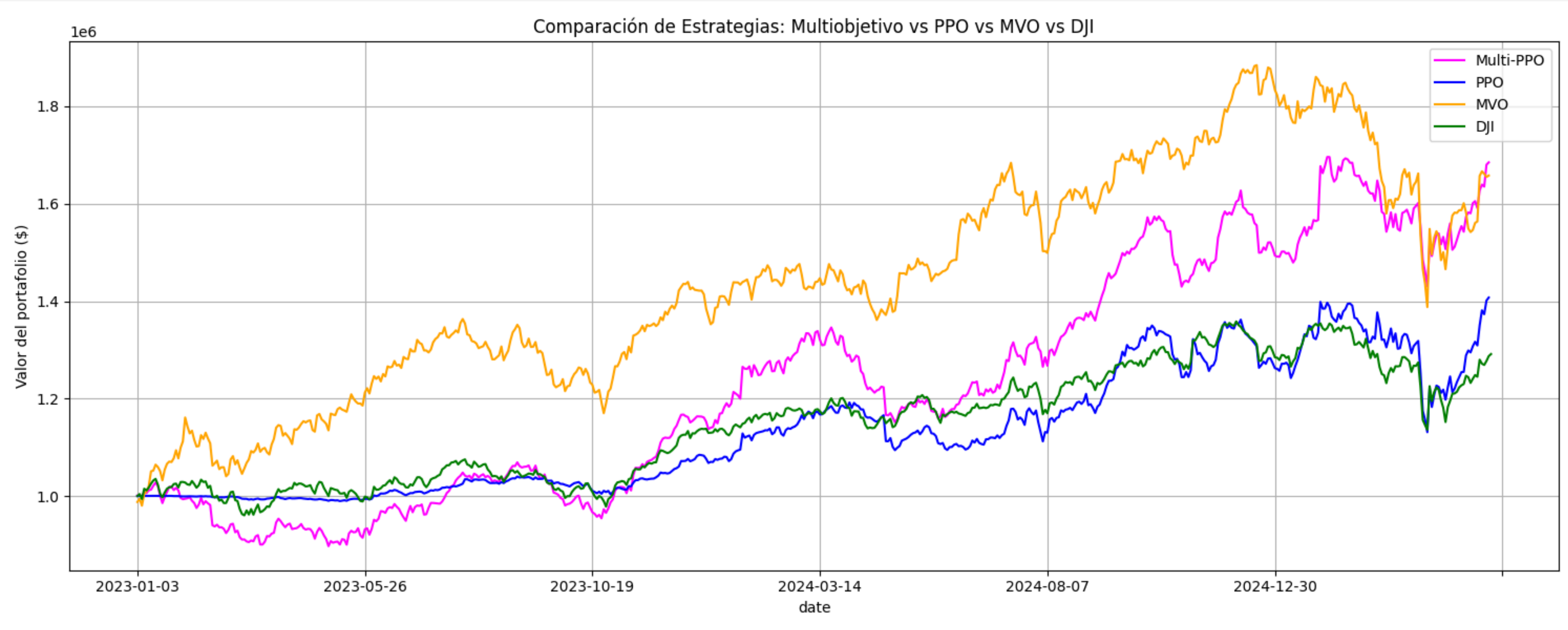
- Use **NSGA-III** to evolve preference weights (λ 's) and find Pareto-optimal strategies.
- Evaluate policies on out-of-sample market data.
- Compare results with **PPO**, **Mean-Variance Optimization** (MVO), and **Dow Jones** (DJI).
- Visualize the **Pareto front** and interpret trade-offs across objectives.

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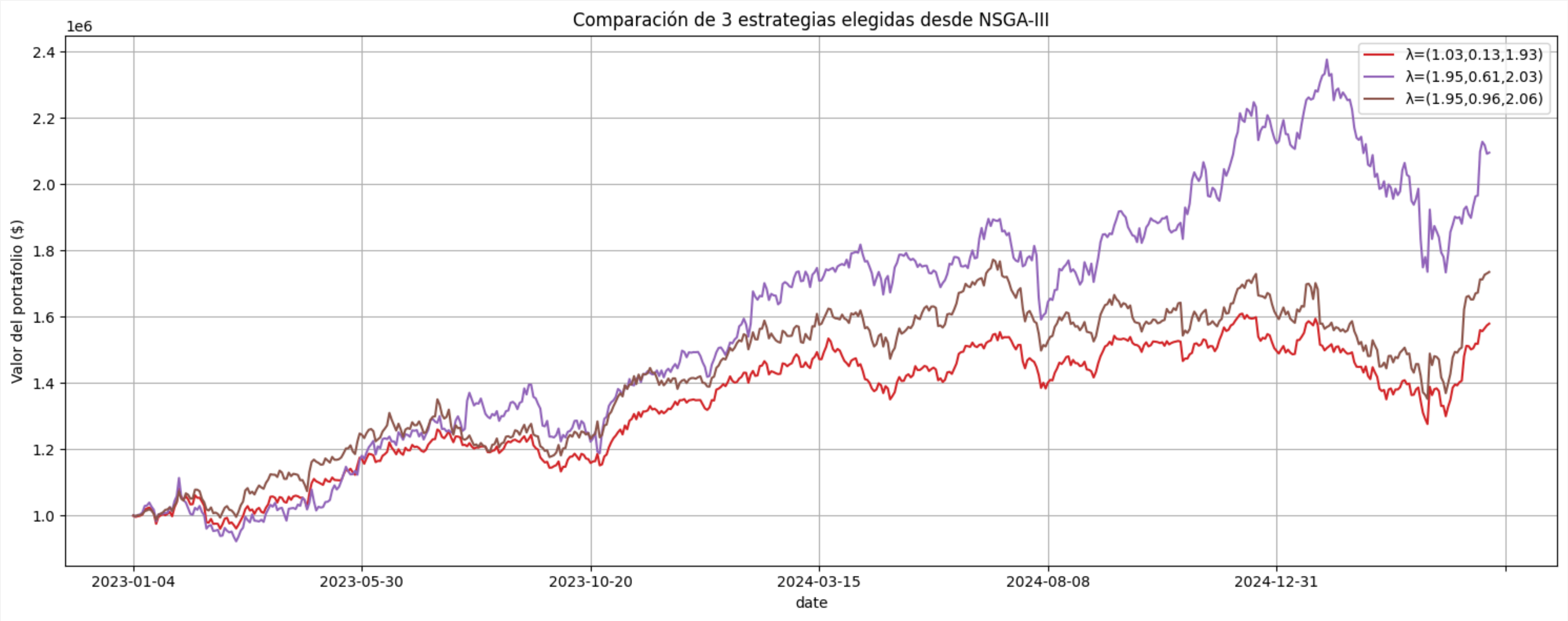
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RESULTS

COMPARATION
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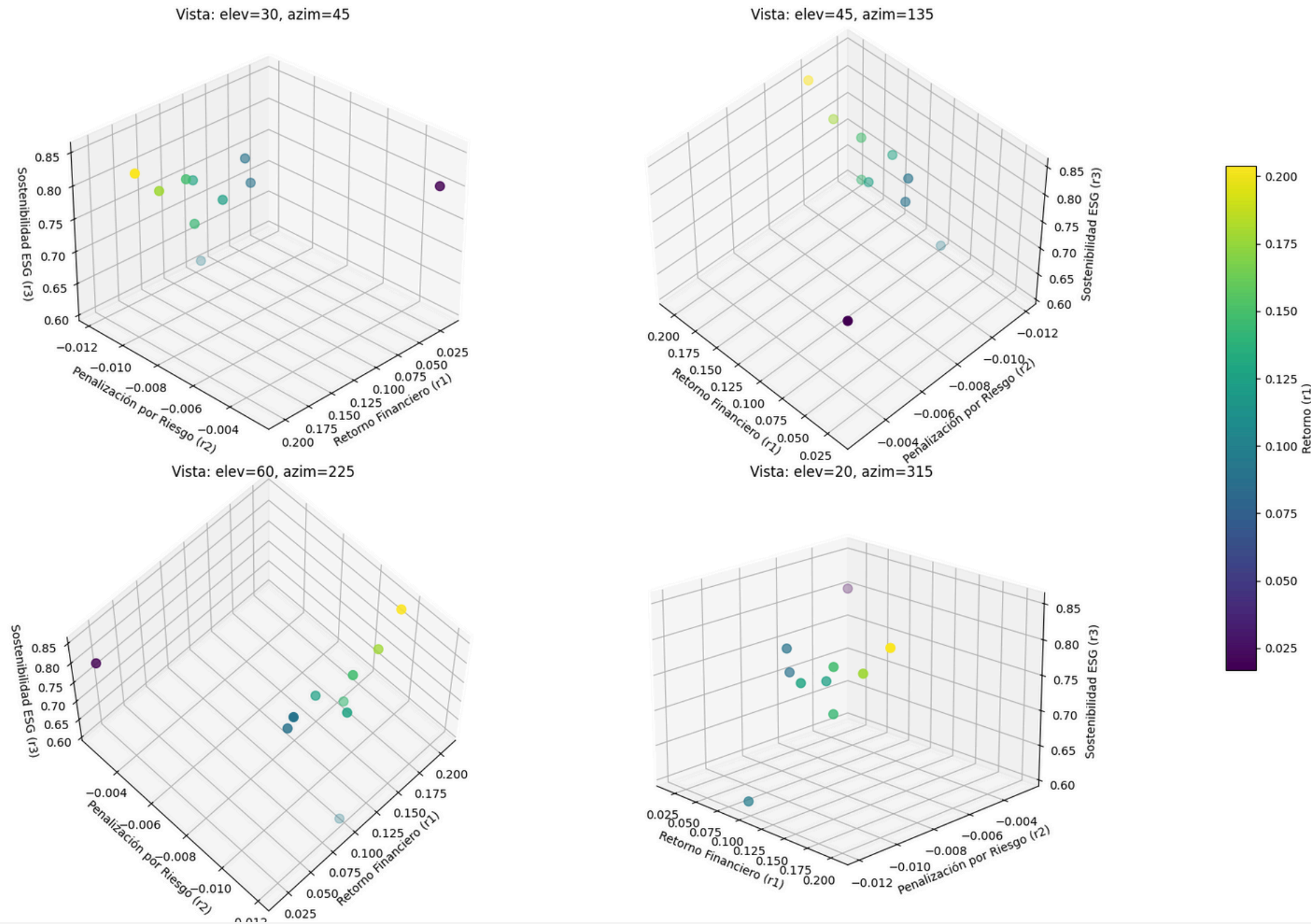


NSGA-III
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PARETO FRONTS
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Frente de Pareto Encontrado por NSGA-III (Múltiples Vistas)



CONCLUSIONS

- This project demonstrated that multi-objective reinforcement learning can be successfully integrated into FinRL to train agents that optimize not only profit, but also risk and ESG sustainability.
- By redesigning the reward function as a vector and applying scalarization, we enabled the agent to learn under diverse investor preferences.

- NSGA-III enabled the discovery of a Pareto front of strategies, each balancing return, risk, and sustainability differently.
- Some of these policies outperformed PPO and MVO, offering better long-term performance, stability, and ethical alignment.

06

THANK YOU

